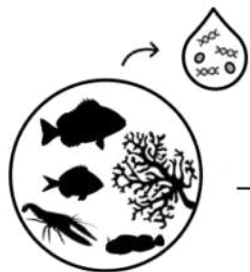


# eDNA quality control

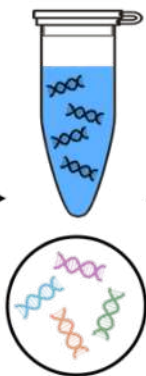
Emilie Boulanger  
DNA-derived data management course  
Santa Marta, Colombia  
22/07/2026

# Quality control Along the workflow from field to data

Field sampling



DNA extraction  
& amplification



High-throughput  
sequencing

```
CTCCAAAT  CCCCTACC
              CCCCGAAC
    CCCCGAAC  CCGCTAGT
    CCCCTACC  CTCCAAAT
CTCCAAAT  CCCCGAAC
    CCGCTAGT  CCCCTACC
    CCGCTAGT  CCGCTAGT
    CCCCGAAC  CCCCTACC
    CCCCTACC  CCGCTAGT
    CCGCTAGT  CTCCAAAT
    CCGCTAGT  CCCCGAAC
CTCCAAAT  CCGCTAGT
```

Bio-informatic  
treatment

```
CTCCAAAT
CTCCAAAT
CTCCAAAT

CCCCGAAC
CCCCGAAC
CCCCGAAC

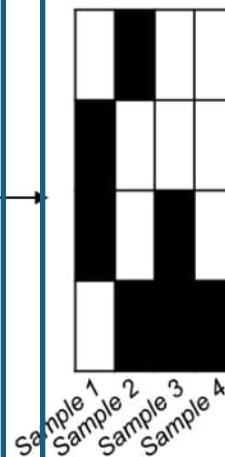
CCCCTACC
CCCCTACC
CCCCTACC

CCGCTAGT
CCGCTAGT
CCGCTAGT
```

Taxonomic  
assignment



Ecological  
dataset



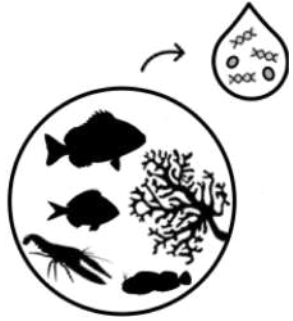
Emilie

Pieter

Elizabeth

# Field sampling

## Replicates and controls



Not much QC to do yet. But make sure to:

- Collect biological replicates (replicate seawater, river water, sediment, ... samples)
  - To minimize false negatives
  - Minimum 3
  - Ideally 6 or more Stauffer et al. 2021
- Collect and filter field blanks: miliQ water
  - To detect & filter false positives

# Reminder

## False positives – false negatives

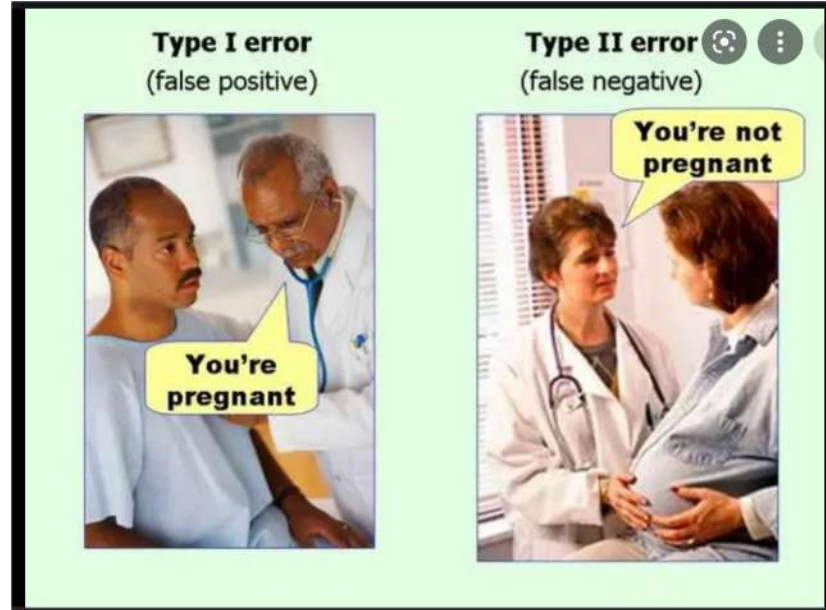
### False positive

A species is detected although it is not present



In eDNA, can come from

- Contamination
- Amplification error
- Chimera
- Sequencing error
- Tag jumping
- Wrong assignments



### False negative

A species is not detected although it is present

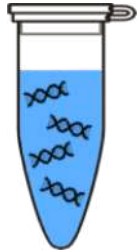


In eDNA, can come from

- Primer bias
- PCR inhibitors
- Undersampling
- Missing references

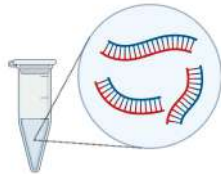
# DNA extraction

## Quantification



- Include extraction blanks
- Is there sufficient DNA for PCR?
  - Measure concentration
- Is your DNA product contaminant-free?
  - Measure absorbance

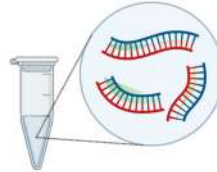
### Quantification of DNA with the Nanospectrophotometer



- Spectrophotometer
- DNA concentration
- Potential RNA and protein contamination.



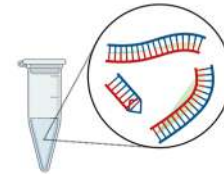
### Quantification of DNA with the Qubit Fluorometer



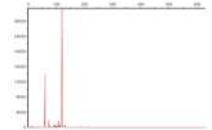
- Fluorometer
- Accurate DNA concentration
- High sensitivity with fluorescent dye



### Quantification and Size of DNA with the TapeStation Gel Electrophoresis System

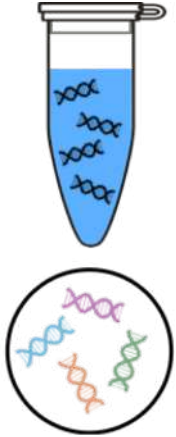


- Gel electrophoresis
- Accurate DNA sizing
- High sensitivity with fluorescent dye

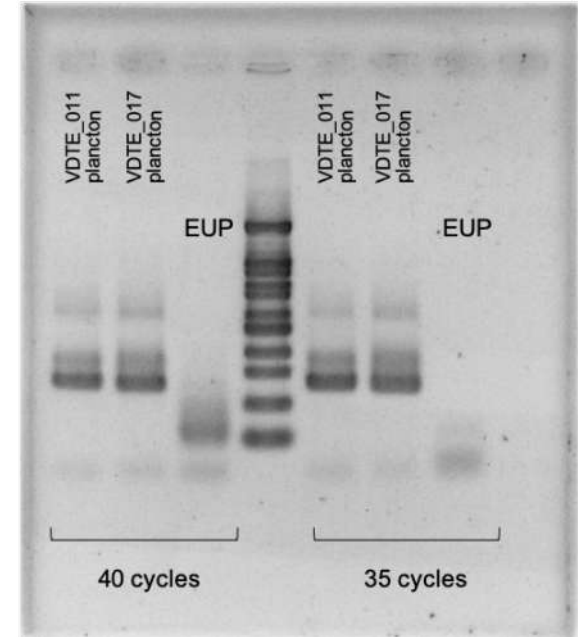
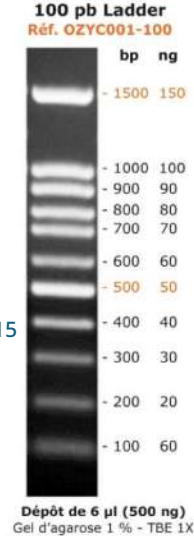


# Amplification – PCR

## Gel electrophoresis

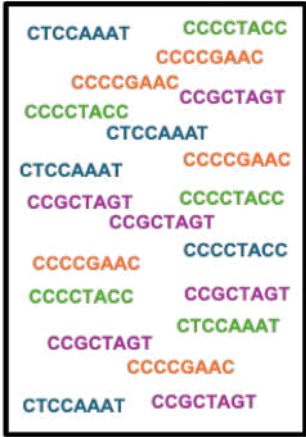


- Include PCR blanks
- Amplify PCR replicates (technical replicates)
  - Minimum 3, ideally 8-12 Ficetola et al. 2015
- Is your amplicon the right size?
- Are your PCR blanks empty?



## Sequencing & bioinformatics

- Do you have sufficient read count per sample?
- Are your reads of good quality?
  - Look within your fastq file at base quality score



FASTQ file sample:

```
@SRR6407486.1 1 length=100
CCTCGTCTACAGCGACAACGTCAGACCCGCAACGGGTGATCGGGCCCTGGGCAAAACGGTTCACCCGGATCTGCCCGATTGACCTACGTCGAAGTG
+SRR6407486.1 1 length=100
BBBBBFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF<FFFFFFFFFFFFFFFFBFFFFFFFFFFFFFFFFBFFFFFFFFFFFF7FFF<F
```

```
@SRR6407486.1 1 length=100
```

CCTCGTCTACAGCGACAAC ... GATTTGACCTACGTCGAAGTG

+SRR6407486.1 1 length=100

BBBBBEEEEEEEEEEEEEEEEEE

\_\_\_\_\_

Base: T  
Quality: 7

Sequence name

**DNA sequence**

Quality line break

### Quality scores

FastQC v0.11.5

File Help

bad\_sequence.txt | good\_sequence\_short.txt

- Basic Statistics
- Per base sequence quality
- Per sequence quality scores
- Per base sequence content
- Per base GC content
- Per sequence GC content
- Per base N content
- Sequence Length Distribution
- Sequence Duplication Levels
- Overrepresented sequences
- Kmer Content

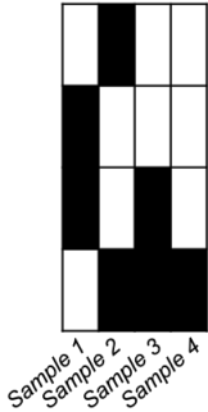
Quality scores across all bases (Illumina v1.3 encoding)

Position in read (bp)

- Trim off low quality ends
- Filter whole reads
- Integrate into error model
- Report % of reads passing quality filter

# Ecological dataset

## Negative controls



- Curate taxonomic assignments
- Compare your true samples with negative controls
  - Subtract contaminants from samples (either full or abundance-based)
- Filter on minimum read-count or relative-abundance threshold

